Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
PI Name:	Wong, Joyce Ph.D.		
Project Title:	Novel Microstructures for Vascular Tissue Engineering		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biotechnology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jywong@bu.edu	**Fax:**	FY 617-353-6766
PI Organization Type:	UNIVERSITY	**Phone:**	617-353-2374
Organization Name:	Boston University		
PI Address 1:	Biomedical Engineering		
PI Address 2:	44 Cummington Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215-2407	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-08-B
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Desai, Tejal (Boston University)		
Grant/Contract No.:	NNJ04HC73G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The major goal of this research project is to tissue-engineer small diameter blood vessel substitutes. Such tissue-engineered blood vessels can also serve as tissue models for studying the effects of microgravity environmental changes on tissue adaptation and to assess potential therapeutic strategies. Requirements for a blood vessel substitute include sufficient burst pressure, viscoelastic mechanical properties, vasoactivity, and nonthrombogenicity. Native blood vessels have evolved a sophisticated structure and hierarchical organization of cell types and matrix molecules in order to satisfy these requirements. Our central hypothesis is that the physiological spatial arrangement of cells and extracellular matrix in an engineered blood vessel will affect the viscoelastic mechanical properties, which in turn, will affect vascular smooth muscle and endothelial cell response. This hypothesis will be tested by answering questions directly posed in Specific Aims 1 and 2 in which the effects of two main biomaterial scaffold properties on tissue function will be explored: (1) microstructure and (2) mechanics. Our strategy of fabricating novel scaffolds with defined microstructure and mechanical properties will help to reveal the effects of specific ultrastructural and mechanical factors		

	on cell response and tissue development. We will use this approach to develop a new generation of biomaterials for vascular tissue engineering that will provide an in vitro model tissue in which parameters can be varied systematically to investigate cell mechanics, tissue development, cell-substrate and cell-cell interactions in controlled microgravity environments in order to attain a better understanding of effects of microgravity on human physiology. Furthermore, we recognize that these tissues may also be used as potential blood vessel replacements that are in high demand and are an integral part of more complex organs and tissues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The ultimate goal of this research project is to create tissue-engineered small diameter vessels that can serve both as vessel replacements and as a model system that could be used to examine the physiological changes that occur during microgravity changes in space flight. We have investigated a number of different polymeric scaffold systems: PDMS (poly dimethylsiloxane), PLGA (poly lactic-co-glycolic acid), and PCL (poly caprolactone). While each of these polymer systems has been used as potential vascular graft materials, there has been little focus on control of the microstructure of these materials. Given that native blood vessels have a highly ordered microstructure, we chose these three polymer systems to investigate the role of microtexture on controlling vascular cell response. We have determined the cellular response of vascular smooth muscle cells (VSMCs) and endothelial cells (ECs) to our microtextured substrata. VSMCs and ECs are the major cellular components of native blood vessels: VSMCs comprise the inner medial layer, and ECs line the lumen of the vessel. In particular, the VSMCs are organized in what is thought to be a helical arrangement. In order to recapitulate this arrangement, we used microtextured substrata to control the orientation and alignment of VSMCs. We used various techniques to assess the alignment, morphology, cytoskeletal organization, phenotypic markers, extracellular matrix organization. In addition, it is critical to have an intact endothelial layer to limit thrombosis. We therefore also assessed the role of substrate microtexture on EC morphology, alignment, and attachment strength.
Bibliography Type:	Description: (Last Updated: 01/09/2014)
Abstracts for Journals and Proceedings	Chown, M.; Sarkar, S.; Desai, T. "Effects of Microtopography on Vascular Smooth Muscle Cells in Biodegradable Tissue Scaffolds" Biomedical Engineering Society Annual Fall Meeting Annals of Biomedical Engineering Supplement , Oct-2004
Abstracts for Journals and Proceedings	Sarkar, S.; Rourke, P.; Desai, T.A.; Wong, J.Y. "Vascular tissue engineering: effect of scaffold microtopography on vascular smooth muscle cell response" 7th World Biomaterials Congress, Sydney AUSTRALIA Transactions of the 7th World Biomaterials Congress , May-2004
Abstracts for Journals and Proceedings	Sarkar, S.; Rourke, P.; Desai, T.A.; Wong, J.Y. "Vascular tissue engineering: Effects of scaffold architecture on smooth muscle cell response" Materials Research Society Annual Meeting, Boston MA Materials Research Society Annual Fall Meeting Book of Abstracts , Dec-2003
Abstracts for Journals and Proceedings	Sarkar, S.; Rourke, P.; Desai, T.A.; Wong, J.Y. "Effects of smooth muscle cell orientation on tissue construct properties" Biomedical Engineering Society Annual Meeting, Nashville TN Annals of Biomedical Engineering Supplement , Oct-2003
Articles in Peer-reviewed Journals	Sarkar, S. S. Sarkar, M. Dadhania, P. Rourke, T.A. Desai, and J.Y. Wong "Vascular Tissue Engineering: Microtextured Scaffold Templates to Control Organization of Vascular Smooth Muscle Cells and Extracellular Matrix" Acta Biomaterialia , Dec-2004
Dissertations and Theses	Rourke, P. "Microtextured Biodegradable Thin Films for Vascular Tissue Engineering" M.S. Thesis, Boston University May-2004
Patents	Provisional Patent filed Sep-2004 Wong, J.Y.; Desai, T.A. "Novel Microstructures for Vascular Tissue Engineering"
Presentation	Chown, M.; Sarkar, S.; Desai, T.A. "Effects of Microtopography on Vascular Smooth Muscle Cells in Biodegradable Tissue Scaffolds" Biomedical Engineering Society Annual Fall Meeting, oral presentation Oct-2004
Presentation	Sarkar, S.; Rourke, P.; Desai, T.A.; Wong, J.Y. "Vascular tissue engineering: effect of scaffold microtopography on vascular smooth muscle cell response" 7th World Biomaterials Congress, May 2004, Sydney, Australia, oral presentation May-2004
Presentation	Sarkar, S.; Rourke, P.; Desai, T.A.; Wong, J.Y. "Vascular tissue engineering: Effects of scaffold architecture on smooth muscle cell response" Materials Research Society Annual Fall Meeting, December 2003, Boston, MA, oral presentation Dec-2003
Presentation	Sarkar, S.; Rourke, P.; Desai, T.A.; Wong, J.Y. "Effects of smooth muscle cell orientation on tissue construct properties" Biomedical Engineering Society Annual Fall Meeting, poster Oct-2003